

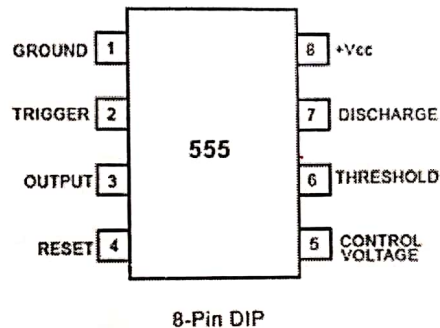
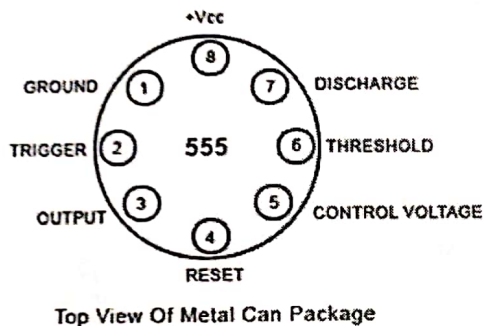
555 Timer IC

The 555 Timer IC got its name from the three 5KΩ resistors that are used in its voltage divider network. This IC is useful for generating accurate time delays and oscillations. The 555 timer IC was introduced in the year 1970 by Signetic Corporation and gave the name **SE/NE 555 timer**. Apart from its applications as a **monostable multivibrator** and **astable multivibrator**, a 555 timer can also be used in **dc-dc converters**, digital logic probes, **waveform generators**, analog frequency meters and tachometers, **temperature measurement** and control devices, **voltage regulators** etc. The **SE 555** can be used for temperature ranges between -55°C to 125° . The **NE 555** can be used for a temperature range between 0° to 70°C .

The 555 Timer IC is available as an 8-pin metal can, an 8-pin mini DIP (dual-in-package) or a 14-pin DIP. The pin configuration is shown in the figures.

This IC consists of 23 transistors, 2 diodes and 16 resistors. The use of each pin in the IC is explained below.

555 TIMER IC



Description of Pins

Pin 1: Grounded Terminal: All the voltages are measured with respect to the Ground terminal.

Pin 2: Trigger Terminal: The trigger pin is used to feed the trigger input when the 555 IC is set up as a monostable multivibrator. This pin is an inverting input of a **comparator** and is responsible for the transition of **flip-flop** from set to reset.

Pin 3: Output Terminal: Output of the timer is available at this pin. There are two ways in which a load can be connected to the output terminal. One way is to connect between output pin (pin 3) and ground pin (pin 1) or between pin 3 and supply pin (pin 8).

Pin 4: Reset Terminal: Whenever the timer IC is to be reset or disabled, a negative pulse is applied to pin 4, and thus is named as reset terminal.

Pin 5: Control Voltage Terminal: The threshold and trigger levels are controlled using this pin. The pulse width of the output waveform is determined by connecting a POT or bringing in an external voltage to this pin.

Pin 6: Threshold Terminal: This is the non-inverting input terminal of comparator 1, which compares the voltage applied to the terminal with a reference voltage of $\frac{2}{3} V_{cc}$.

Pin 7 : Discharge Terminal: This pin is connected internally to the collector of transistor and mostly a capacitor is connected between this terminal and ground. It is called discharge terminal because when transistor saturates, capacitor discharges through the transistor. When the transistor is cut-off, the capacitor charges at a rate determined by the external resistor and capacitor.

Pin 8: Supply Terminal: A supply voltage of + 5 V to + 18 V is applied to this terminal with respect to ground (pin 1).

The 555 IC has the following operating modes:

- **Astable (free-running) mode** – The 555 can operate as an electronic oscillator.
An Astable Circuit has no stable state - hence the name "astable". The output continually switches state between high and low without any intervention from the user, called a 'square' wave. This type of circuit could be used to give a mechanism intermittent motion by switching a motor on and off at regular intervals. It can also be used to flash lamps and LEDs, and is useful as a 'clock' pulse for other digital ICs and circuits....
- **Monostable (one-shot) mode** – In this mode, the 555 functions as a "one-shot" pulse generator. A Monostable Circuit produces one pulse of a set length in response to a trigger input such as a push button. The output of the circuit stays in the low state until there is a trigger input, hence the name "monostable" meaning "one stable state". This type of circuit is ideal for use in a "push to operate" system for a model displayed at exhibitions. A visitor can push a button to start a model's mechanism moving, and the mechanism will automatically switch off after a set time....
- **Bistable (flip-flop) mode** – The 555 operates as an SR flip-flop. A Bistable Mode or what is sometimes called a Schmitt Trigger, has two stable states, high and low. Taking the Trigger input low makes the output of the circuit go into the high state. Taking the Reset input low makes the output of the circuit go into the low state.

14.16.1 Waveform Generator *(555 timer in Astable mode)*

The monolithic integrated circuit 555 timer can be used as square-wave generator, linear saw-tooth generator, pulse generator and time delay generator. The 555 timer shown in Fig. 14.42. The capacitor C_1 charges through the resistances R_1 and R_2 and discharges through the resistance R_2 only. The duty cycle can be controlled by the ratio of resistances R_1 and R_2 . Here, the capacitor charges and discharges between $(1/3)V_{CC}$ and $(2/3)V_{CC}$. The charging and discharging times are independent of supply voltage.

The charging time of the capacitor is

$$t_1 = 0.693 (R_1 + R_2) C_1$$

The discharging time of the capacitor is

$$t_2 = 0.693 R_2 C_1$$

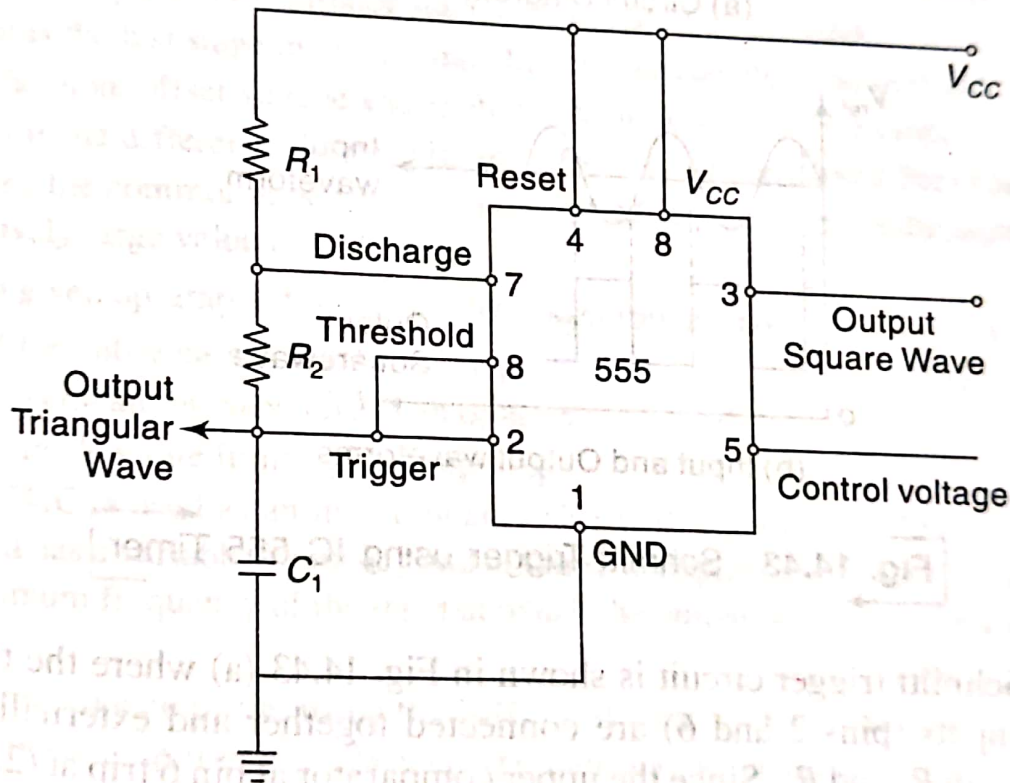


Fig. 14.42 Square and triangular waveforms generator using 555 Timer

Therefore, the time period of the waveform is given by

$$T_1 = t_1 + t_2 = 0.693 (R_1 + 2R_2)C_1$$

The frequency of oscillation is given by

$$f = 1/T = 1.44/C_1 (R_1 + 2R_2)$$

where R_1 and R_2 are in ohms and C_1 is in Farads.

The 555 timer can also be used for the construction of Monostable multivibrator and Schmitt trigger.